Do Leaders Export Pollution While Importing Wealth?

Evidence of "Green Favoritism" in Autocratic Political Economies

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March 8, 2026

Research Question: Do political leaders systematically direct clean, high-value economic development to their birth regions while diverting pollution-intensive industry to peripheral areas?

Abstract

Standard political economy literature has robustly established that leaders direct economic resources toward their birth regions, a pattern detectable via satellite nighttime lights (Hodler and Raschky 2014; Bora 2025). However, this literature focuses exclusively on the *quantity* of development, overlooking its environmental dimension. We propose studying a novel phenomenon we term "Green Favoritism": the hypothesis that autocratic leaders practice a NIMBY strategy, channeling clean sectors to home regions while relegating pollution-intensive industry to peripheral areas. Using NASA's Ozone Monitoring Instrument (OMI) to measure tropospheric NO_2 as a proxy for industrial pollution and nighttime lights as a proxy for economic activity, we test whether birth regions exhibit a *decoupling* effect: rising output without a proportionate rise in pollution. Such a finding would constitute the first evidence of systematic inequality in the *quality* of development—not just its quantity—with direct implications for environmental justice in autocratic regimes.

Data

We combine satellite-derived and political data across up to 200 countries:

- **Political Data:** The *Political Leaders' Affiliation Database (PLAD)* (Bomprezzi et al. 2025) provides subnational birth region data for heads of state and government, enabling us to identify treatment (birth region) and control (non-birth regions) within each country-year.

- **Industrial Pollution (NO_2):** Annual global surface NO_2 estimates from the Atmospheric Composition Analysis Group (Cooper et al. 2022), combining NASA's Ozone Monitoring Instrument (OMI) with the GEOS-Chem chemical transport model at 0.01° resolution (2005–2019). NO_2 is a well-validated proxy for fossil-fuel combustion and heavy industrial activity, making it an ideal measure of pollution-intensive economic development.

- **Economic Activity (Nighttime Lights):** We use a harmonized panel combining *DMSP-OLS* (1992–2013) and *VIIRS* (2012–Present) (Elvidge et al. 2013), well-established proxies for subnational economic output. The 2012–2013 overlap period enables inter-calibration of the two sensors.

- **Administrative Boundaries:** GADM (Global Administrative Map) shapefiles at ADM2 resolution (Global Administrative Areas 2022), used to spatially aggregate all raster data to consistent subnational units and match them to PLAD's birth region identifiers.

All data are extracted and aggregated to ADM2 level, then matched to PLAD records by country, region, and year. The analysis panel runs from **2005 to 2019**, constrained by the availability of the ACAG NO_2 estimates.

Methods

To isolate the causal impact of political power on environmental quality, we employ a **Difference-in-Differences (DiD)** framework with high-dimensional fixed effects. The identifying variation comes from leader transitions: a birth region switches from "untreated" to "treated" when its leader ascends to power. Our primary regression specification is:

$$Y_{it} = \beta\,(\text{BirthRegion}_i \times \text{InPower}_{ct}) + \alpha_i + \gamma_{ct} + \varepsilon_{it}$$

where Y_{it} is log nighttime lights or log NO_2; BirthRegion_i is a birth-region indicator; InPower_{ct} indicates power in country c at time t; α_i are region fixed effects; and γ_{ct} are country-year fixed effects absorbing national shocks. β captures the birth-region treatment effect relative to all other regions in the same country-year.

We estimate this specification separately for nighttime lights and NO_2. The central test of "Green Favoritism" is whether $\hat{\beta}^{\text{lights}} > 0$ while $\hat{\beta}^{NO_2} \leq 0$—or equivalently, whether the implied **pollution elasticity of growth** (the ratio $\hat{\beta}^{NO_2}/\hat{\beta}^{\text{lights}}$) is significantly lower for birth regions than for control regions. We will also estimate **event-study** specifications, replacing the treatment indicator with a set of leads and lags around leader ascension, to assess pre-trends and the temporal dynamics of treatment effects.

Feasibility: All data are publicly available. We have already accessed the PLAD dataset, downloaded GADM shapefiles, and built a fully reproducible ACAG ingestion pipeline that downloads, checkpoints, and aggregates NO_2 to ADM2. Preliminary data merging has been completed, confirming that the datasets can be matched at the ADM2 level.

Preliminary Findings (March 2026)

Using the replication-style long DMSP sample (1992–2013), we recover a positive birth-region nightlights effect consistent with the literature ($\hat{\beta} \approx 0.0128$, $p = 0.030$). However, in the restricted analysis window required by ACAG NO_2 availability (2005–2019), the baseline birth-region effect in nightlights is close to zero and statistically insignificant.

In pooled 2005–2019 specifications, we do not find evidence for the intended "Green Favoritism" pattern. The contemporaneous NO_2 effect is weakly positive, and a direct pollution-intensity outcome $\ln(NO_2) - \ln(\text{nightlights})$ is positive at lag 0 (marginal significance), indicating no support for cleaner growth in birth regions in this restricted sample.

These results are preliminary and motivate a heterogeneity-focused next step (especially autocracy-focused subsamples/interactions), where theory predicts stronger favoritism.

Related Literature

Our project builds on a rich empirical literature on regional favoritism. Hodler and Raschky (2014) provide the foundational evidence that leaders direct economic resources to their birth regions across 126 countries using nighttime lights. Bora (2025) replicates and extends this result with modern satellite data, showing the pattern is robust across crisis periods. Favoritism has also been documented beyond birth regions: Bomprezzi et al. (2024) show that donors direct more aid to the birth regions of leaders' spouses, illustrating how informal political influence operates through multiple ties.

Our project departs from all prior work by introducing the *environmental* dimension of favoritism. Whereas existing studies measure only the quantity of economic activity in favored regions, we test whether the *composition* of development systematically differs—clean versus dirty sectors—between politically favored and disfavored regions. This connects to the "Pollution Haven Hypothesis" (Copeland and Taylor 2004) in environmental economics, which predicts that areas with weaker regulatory oversight attract more pollution-intensive industries. We apply this logic *within* countries, testing whether autocrats informally enforce a domestic pollution haven regime by concentrating dirty industry in politically marginal areas while preserving amenity quality in their home regions.

Key Figure

Figure 1 presents a schematic of our main result: a two-panel event-study plot centered on leader ascension ($t = 0$). **Panel A** shows nighttime lights rising significantly in birth regions post-ascension with flat pre-trends, replicating the favoritism finding. **Panel B** shows the "Green Favoritism" signature: NO_2 remains flat in birth regions while rising in peripheral control regions, despite their economic gains—the core decoupling result.

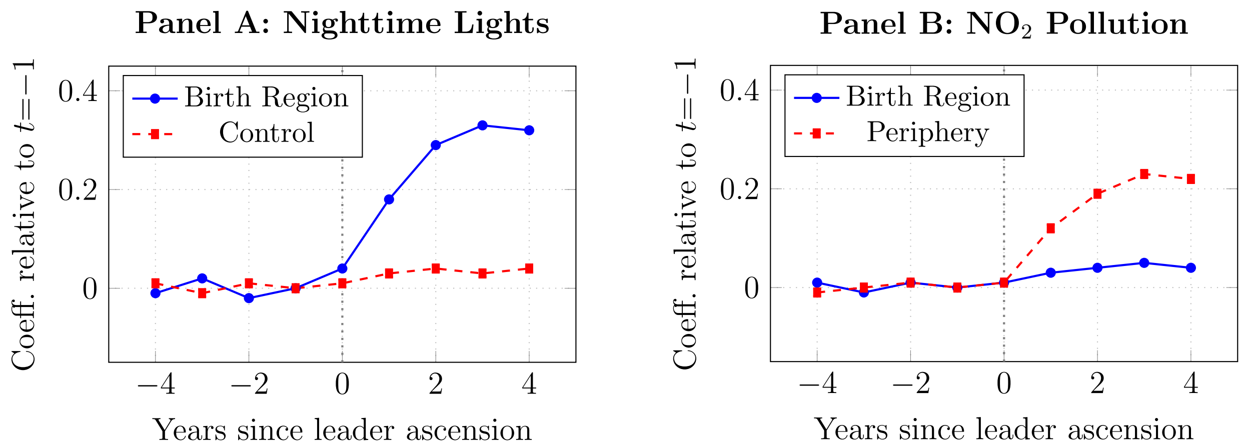

Figure 1: *Schematic of expected results* (illustrative values). Vertical dotted line marks leader ascension. Panel A replicates the favoritism finding: birth regions see a positive break in nighttime lights with flat pre-trends. Panel B shows the "Green Favoritism" signature: NO_2 rises in peripheral regions but not in birth regions, despite their economic gains.

Division of Labor

Richard Schulz brings experience with geospatial data engineering and reproducible satellite-data pipelines; Muhammad Altaf has econometrics training in Stata/R; Ruby Wang has a background in environmental data analysis.

- Data acquisition and processing (ACAG NO_2 pipeline, GADM matching, panel construction): *Richard Schulz*
- Econometric analysis and regression implementation: *Muhammad Altaf*
- Literature review and related work write-up: *Muhammad Altaf*
- Robustness checks and heterogeneity analyses: *Ruby Wang*
- Interpretation of results and final paper draft: Richard Schulz, Ruby Wang, Muhammad Altaf

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